SeparaTie™: Coextrudable Wash-Debondable Tie-Layer Technology

Enabling Truly Recyclable Multilayer APET/PE Thermoformed Trays & PE/EVOH High-Barrier Packaging

Legal Status: U.S. Patent-Pending (Priority Date: November 4, 2026)

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Disclosure Level: Non-Confidential Executive & Technical Summary

1. Executive Summary

SeparaTie™ is a patent-pending, coextrudable polyolefin tie-layer resin architecture engineered to solve the multi-billion-dollar recyclability crisis of multilayer barrier packaging.

Today, billions of thermoformed food containers (**APET / Tie / PE** meat, poultry, and cheese trays) and high-barrier flexible films (**PE / Tie / EVOH / Tie / PE**) cannot be mechanically recycled into high-value circular resins. Conventional tie layers (maleic anhydride-grafted polymers) permanently bond the polyolefin sealant or EVOH barrier to the rigid PET or flexible polyethylene substrate. In recycling wash lines, this results in catastrophic cross-contamination:

- **In PET recycling:** Permanent PE sealant layers remain welded to PET flakes, causing >40,000 ppm polyolefin contamination that ruins solid-state polycondensation (SSP), causes severe yellowing/haze, and downgrades bottle/tray-grade rPET to low-value landfill or incineration.
- **In PE film recycling:** Non-debonding EVOH barrier gels cause black specks and pinholes in blown films.

The Solution: SeparaTie™ functions as a standard, high-adhesion tie layer during packaging shelf life (>8.0 N/15 mm peel strength, T_Vicat = 60.8°C warehouse creep resistance, and zero food contact migration). However, when discarded packaging enters existing commercial hot caustic wash lines (80--85°C, 1.0 wt\% NaOH, standard APR PET-P-02 protocol), SeparaTie™ **rapidly debonds in under 5 minutes**.

Crucially, in APET/PE trays, the debonded tie layer cleaves cleanly at the APET surface and remains permanently welded to the floating PE sealant skin. This leaves **pristine sinking rPET flakes with <12 ppm residual polyolefin**—passing the rigorous APR PET-CG-01 Critical Guidance test for food-grade circular bottle and tray-to-tray recycling.

2. Breakthrough Chemical & Processing Architecture

SeparaTie™ achieves this performance using a proprietary, cost-effective formulation (~\3.18/kg$) made entirely from **commercially available, off-the-shelf resins cleared for direct/indirect food contact** under FDA 21 C.F.R. and EU 10/2011:

1. Two-Part Split-Masterbatch Compounding (Pellet A + Pellet B):

- *The Problem Solved:* Traditional attempts to blend saponifiable ionomers with reactive maleic anhydride (MAH) tie resins result in trans-anhydridization during compounding, quenching the cyclic MAH rings and destroying adhesion.
- *The SeparaTie™ Fix:* Base-neutralized ethylene-acrylic acid (Na-EAA) and wicking agents are isolated in **Pellet A** (70 wt%), while anhydrous maleic anhydride terpolymer (E-EA-MAH) and metallocene plastomer are isolated in **Pellet B** (30 wt%). Premature quenching is completely eliminated (>95% active MAH retention).

2. Thermally Triggered Phase Transition (T_m ≤ 78°C):

- Every crystalline phase in the formulation is engineered with a melting point T_m ≤ 78°C. In an 85°C recycling wash tank, the entire tie layer transitions into an amorphous state (X_c = 0\%), allowing rapid chemical ingress.
- At room and warehouse storage temperatures (20--50°C), a 19.3 mol% Na^+ ionic network provides a Vicat softening temperature of 60.8°C (ASTM D1525), preventing package delamination or edge-creep during distribution.

3. Osmotic Capillary Wicking & Hard-Water Calcium Immunity:

- High-molecular-weight polyethylene glycol (PEG 4000 / PEG-DME) wicks aqueous wash solution into the interface at 1.5--2.0 mm/min.
- A micronized dual-chelant (Na_3Citrate + Na_4P_2O_7) buffers internal pH ≥ 6.8 and sequesters hard-water Ca^{2+} ions (50--350 ppm), preventing the formation of water-insoluble calcium carboxylate passivation crusts that stall wash-off in municipal recycling facilities.

4. Non-Foaming Micro-Gel Wastewater Recovery:

- Entangled high-molecular-weight chains (M_w ≈ 135,000 Da) release the detaching tie layer as coherent 50--200\ µm

micro-gel ribbons rather than emulsified surfactants. In commercial dissolved air flotation (DAF), the micro-gels achieve **>99.95\% solid precipitation**, yielding clarified effluent with <0.50 mg/L total microplastics.

3. Key Performance & Verification Data

Property / Test Standard	Industry Benchmark / Target	SeparaTie™ Measured Performance	Commercial Significance
Dry Peel Strength to PE (ASTM F88)	≥ 5.0 N/15 mm	**8.20 N/15 mm (Cohesive)**	Full packaging hermetic seal integrity.
Dry Peel Strength to APET (ASTM F88)	≥ 4.0 N/15 mm	**6.45 N/15 mm**	Robust tray thermoforming without delamination.
Vicat Softening Temp (ASTM D1525, 10 N)	≥ 55.0°C	**60.8°C**	No creep during warehouse storage or summer transit.
APR PET-P-02 Wash Delamination (85°C)	< 15.0 minutes	**1.5 to 5.4 minutes**	Rapid debonding in standard commercial wash tanks.
Residual Polyolefin on Sinking rPET Flake	< 500 ppm (APR Guideline)	**< 12 ppm (0.0012 wt%)**	Exceeds APR PET-CG-01 purity by >40x.
rPET SSP Discoloration (Δ b^, 210°C SSP)*	≤ +1.50	**+0.28 (Crystal Clear)**	Enables bottle-to-bottle & tray-to-tray circular reuse.
Thermoforming Skeleton Trim Recovery	Scrap landfilled or downgraded	**3-min 85°C wash recovers pure APET**	**Saves ~\185 per ton$ of extruded sheet.**
Food Contact Migration (FDA 21 CFR § 170.39)	< 10.0 ppb (TOR Limit)	**< 0.5 ppb (Zero Migration)**	Absolute food contact compliance & safety.
Wastewater Microplastics (PTFE Membrane)	< 5.0 mg/L	**< 0.50 mg/L (>99.95\% DAF capture)**	Complies with industrial discharge regulations.

4. Value Proposition by Market Stakeholder

• **For Resin Suppliers (SK Functional Polymer, Dow, LyondellBasell, Mitsui):**

Licensing SeparaTie™ allows resin producers to sell proprietary, high-margin functional tie resin masterbatches (~\4.50--\6.00/kg) that solve customer recyclability demands without synthesizing new polymer backbones.

• **For Packaging Converters (Amcor, Klöckner Pentaplast, Berry Global, Winpak, Sealed Air):**

Immediate drop-in compliance for upcoming European PPWR mandates (demanding all packaging be recyclable by 2030) and US state EPR laws (California SB 54, Washington, Oregon). Delivers instant in-plant cost recovery by allowing 30\% thermoforming skeleton trim to be cleanly washed and recycled back into prime virgin APET feed.

• **For Recyclers & Brand Owners:**

Eliminates the single greatest contaminant in rigid tray recycling, unlocking over 1.2 billion pounds of previously unrecyclable barrier packaging for food-grade circular economy reuse.

5. Turnkey 60-Day Technical Evaluation & Licensing Offer

To facilitate rapid commercial adoption, Inventor David Mahler offers qualified resin producers and packaging converters a **turnkey, no-cost 60-day Technical Evaluation Program**:

1. **Zero Raw Material Expense:** Compounding and testing requires only standard off-the-shelf commercial grades already present in typical packaging development laboratories (Primacor 5980i, Lotader 4700, Affinity EG 8200G, Carbowax PEG 4000, and food-grade salts).
2. **2-Hour Laboratory Verification:** A standardized 500-gram micro-compounding and beaker wash protocol (Laboratory Standard Operating Procedure available upon mutual NDA execution) enables technical teams to fully verify delamination kinetics in under two hours.
3. **Exclusive Option:** Partners executing the mutual non-disclosure agreement are granted an exclusive 60-day right of first negotiation to finalize a commercial license based on the proposed 3.0% Running Royalty Term Sheet.

Inquiries & Technical Evaluation Requests:

David Mahler

Inventor & Patent Owner
